- Ensuring security and enforcing access control policies

Orchestration tools are instructed declaratively, such as through YAML files. Following the format and language defined by the specific tool, you usually describe the state you'd like to achieve; for example, you would like to have three replicas of your web application container deployed in a cluster, exposing its services to the internet.

Examples of container orchestrators are Kubernetes (a CNCF project), Docker Swarm, and Apache Mesos. In this book, you'll learn how to use Kubernetes to orchestrate the containers for your Spring applications.

1.6.3 Serverless

After moving from virtual machines to containers, we can abstract the infrastructure even more: that's where serverless technologies are placed. The serverless computing model enables developers to focus on implementing the business logic for their applications.

The name *serverless* might be misleading. Of course there is a server. The difference is that you do not need to manage it or orchestrate the application's deployment on it. That's a platform responsibility now. When you use an orchestrator like Kubernetes, you must still consider infrastructure provisioning, capacity planning, and scaling. In contrast, a serverless platform takes care of setting up the underlying infrastructure needed by the applications, including virtual machines, containers, and dynamic scaling.

Serverless architectures are usually associated with functions, but they comprise two main models that are often used together: